

Intellectual Property

Lecture 2

Copying: Music

- Improved technology allows easy, fast, cheap and widespread copying of music on Web.
- Entrepreneurs create businesses to facilitate storing and sharing of music files.
 - Many individuals set up free sites for music sharing, too.

Copying: Movies & TV Programs

- Improved digital technologies + greater bandwidth = copying and transfer of movies and TV programs
- Some businesses provide(d) free services to facilitate copying of broadcast material

Copying: Software

- Same technologies contributed to unauthorized copying of software
 - commercial (i.e., within or by companies)
 - non-commercial (individuals)
- Many individuals & whole businesses continue to do this
 - Reproducing, transporting, & selling copies of software, manuals and supporting materials.
 - Also covers those who give away such material for free.

Copying: Books

- Again, improved technology enables copying of books that:
 - is simple to perform
 - quick
 - cheap
- Counterfeiters
 - Textbooks, novels, other printed matter
 - Profit is in not paying publishers or authors for their IP
- E-books use encryption to reduce copying
 - But some of these schemes have been cracked.

Solutions to IT's impact on

- There are three groups of solutions
 - Technological
 - Markets and Management
 - Regulations and Enforcement
- Technological
 - Expiration date encoded.
 - “Activation” features.
 - Digital-rights management (DRM) – encryption and other schemes
 - Can prevent saving, printing, making more than N copies, distribution, extracting excerpts, ffw over commercials

Solutions to IT's impact on

- Markets and Management
 - Subscribe to services
 - Collect fees from users and large organizations.
 - Meter usage of copyrighted material on a network.
 - Offer discounts to educational users.
 - Educate the public about the value of copyrighted material produced by creators and publishers.
- Regulations and Enforcement
 - Enforce current laws and punish abusers.

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Patents

- Copyrights:
 - Protect the expression of an idea in a fixed or tangible form.
 - Are cheap, easy to obtain, and last a long time.
 - Allow fair dealing of the intellectual property.
- Patents:
 - Protect new, non-obvious, and useful processes.
 - Are expensive, difficult to obtain, and last for short periods of time (20 years in Canada & USA)
 - Allow licensing to other developers.

Copyright & software forms

- Recall that software can take several forms:
 - source code
 - object code
 - microcode (i.e., within processors)
- Each of these can be copyrighted
 - For instance, copyrighting object code deals with software piracy.

Copyright & software forms

- But what about software versus hardware?
- Software is copyrightable
- Yet we know that the same software could be turned into hardware and hardware falls under the purview of patent.
- – (Note: the hardwired version could not be copyrighted.)

Patents

- Possible patentable material:
 - machines
 - industrial processes
 - articles of medicines
 - manufacture that are novel, useful, and non-obvious
- Patent must describe:
 - how to make the claimed invention
 - how to use it

Copyright vs Patent

<u>Feature</u>	<u>Copyright</u>	<u>Patent</u>
Protects	Creative works	Inventions
Registration	Automatic in many jurisdictions	Required
Duration	Usually life + 50/70 years (BD: generally life + 60)	20 years from filing
International	Berne/WCT/TRIPS	Paris/PCT/TRIPS



Copyright Laws

International: Berne Convention, Universal Copyright Convention, WIPO Copyright Treaty (1996), TRIPS Agreement.

Bangladesh: Copyright Act, 2023.

Patent Laws

International: Paris Convention, Patent Cooperation Treaty (PCT), TRIPS Agreement.

Bangladesh: Patents Act, 2022.

Territorial Nature of IP

Copyright and patents are territorial rights. Copyright is widely recognized internationally through treaties such as the Berne Convention, while patents must generally be filed separately in each country where protection is desired.

Difference Between Trademark and Trade Secret

Trademark

Protects brand identity (name, logo, slogan, etc.)

Usually requires registration for stronger legal protection

Can last indefinitely with renewal and continued use

Publicly visible

Example: McDonald's Golden Arches

Trade Secret

Protects confidential business information

Does not require registration

Lasts as long as the information remains secret

Must remain confidential

Example: Coca-Cola secret formula



Trademark Law: Trademarks Act, 2009 (with Trademarks Rules, 2015).

In Bangladesh, trademarks are registered with the Department of Patents, Designs and Trademarks (DPDT). A trademark is initially protected for **7 years** and can be renewed for successive **10-year** periods.

Trade Secret Law: No specific trade secret statute in Bangladesh; protection is mainly through **contracts (NDAs/confidentiality agreements)**, **breach of confidence**, and **TRIPS obligations**.

Protection continues **as long as the information remains secret** and reasonable measures are taken to maintain its confidentiality. If the information becomes public through lawful means, trade secret protection is lost.

Thank You